

Chapter 11

The Role of the Pinhole in Confocal Imaging Systems

T. WILSON

Department of Engineering Science, University of Oxford, Parks Road, Oxford. OX1 4UU

INTRODUCTION

Confocal scanning microscopes are particularly attractive by virtue of their enhanced lateral resolution, purely coherent image formation and optical sectioning (Wilson and Sheppard, 1984). It is probably the latter property which is most useful as it gives rise to the ability to image a thick specimen in three-dimensions. This is possible because the optical system images information only from a thin region in the neighbourhood of the focal plane. This permits us to store many image slices in a computer to give a three dimensional data set which describes the object. There are now many sophisticated computer software systems available to display this data in various ways. Obvious examples include the extended focus technique (Wilson and Hamilton, 1982) in which we merely add up (integrate) the images from various depths to provide an image of greatly extended depth of field. We may also produce images where object height is coded as brightness or combine the whole data set to provide an isometric view of the object, Figure 1. It is also possible to use false color to label features of interest or, by simple processing, to obtain stereoscopic pairs (van der Voort, 1985). Other forms of image display are discussed in Chapter 10.

Figure 2 shows a schematic of a confocal scanning optical microscope. The only difference between a confocal and a conventional scanning microscope is that the confocal arrangement uses a point detector rather than a large area one. All the considerable advantages of confocal microscopy derive from the small size of this detector. The success or failure of a particular microscope implementation in achieving true confocal operation depends on the correct choice of pinhole size and shape to approximate as closely as possible the ideal point detector.

In many applications, however it is not possible to use as small a detector as we would like because of signal to noise problems. This is particularly true in fluorescent imaging. In the following, therefore, we will discuss the origins of the optical sectioning property and show how it deteriorates as detector geometries of finite size are employed. We will begin by considering brightfield microscopes and conclude with a discussion of fluorescent systems.

THE OPTICAL SECTIONING PROPERTY

The physical origin of the property is shown in Figure 3. When the object is in the focal plane (dashed lines) the reflected light is focussed on to the pinhole and a large signal is detected. On the other hand, when the object is positioned out of the focal plane, a defocussed spot is formed at the pinhole, and the measured intensity is greatly reduced. The strength of the op-

tical sectioning, that is the rate at which the detected intensity falls off with axial distance, is clearly a function of both the detector size and the object feature. If we begin our discussion by considering an ideally small point detector we can readily calculate the images of idealised objects. We will consider the images of a single point, a line and a plane. We show, in Figure 4, these images as a function of a normalised axial distance u , which is related to real axial distance, z , via

$$u = \frac{8\pi}{\lambda} \sin^2(\alpha/2) z \quad (1)$$

where λ is the wavelength and $\sin\alpha$ the objective lens numerical aperture. The results for the case of the point and line are shown for focussing directly on top of the feature. In particular we see that the sectioning is twice as weak in the case of a point object as a plane for $I(u) = 0$ whereas; at the half intensity resolution, $u_{1/2}$, the values are at about $u = 4$ and $u = 3$ respectively. (These benchmark values will be useful in interpreting later graphs where $u_{1/2}$ is a parameter).

In the paraxial approximation we can write the axial image of a plane as

$$I(u) = \left(\frac{\sin u/2}{u/2} \right)^2 \quad (2)$$

We see, therefore, that the strength of the sectioning depends on the size and shape of the specific object feature imaged. In order to have a convenient experimental measure of sectioning we will concentrate on the planar object. In practice we will scan a perfect reflector through focus in a reflection microscope and note the form of the response. We define a metric of the sectioning as being the full-width-half-maximum of this response. Equation (2) is a paraxial expression: fortunately it is possible to derive a high angle equivalent which is more accurate at high numerical aperture, (Sheppard and Wilson, 1981b), and based on this theory it is possible to predict optical sectioning strengths as shown in Figure 5.

THE OPTICAL SECTIONING PROPERTY WITH A FINITE-SIZED CIRCULAR DETECTOR AND COHERENT LIGHT

It is clear from Figure 3 that the strength of the sectioning will become less as the detector size is increased. In the limit of a very large detector the sectioning disappears altogether. Figure 6 illustrates the effect. Here the pinhole size is denoted by a normalised radius, v_p , which is related to real radius, r_p , via

$$v_p = \frac{2\pi}{\lambda} \sin\alpha r_p \quad (3)$$

We see that the curves corresponding to $v_p=0$ and $v_p=1$ are indistinguishable on the scale of Figure 6, which implies that

the strength of the sectioning is relatively insensitive to pinhole size up to some limit (Wilson and Carlini, 1987a). This is borne out in practice in Figure 7 where we see that the response of the microscope is essentially indistinguishable for pinholes of $5\ \mu\text{m}$ and $10\ \mu\text{m}$ diameter. These results were obtained (Wilson

and Carlini, 1988a) using red light ($6328\ \text{\AA}$ wavelength) from a helium neon laser and a 32×0.5 numerical aperture objective lens. The effect of introducing larger detector pinholes is shown in Figures 8(a) and 8(b). It is seen that the curves become distinctly broader by the time a $150\ \mu\text{m}$ diameter aperture is used.

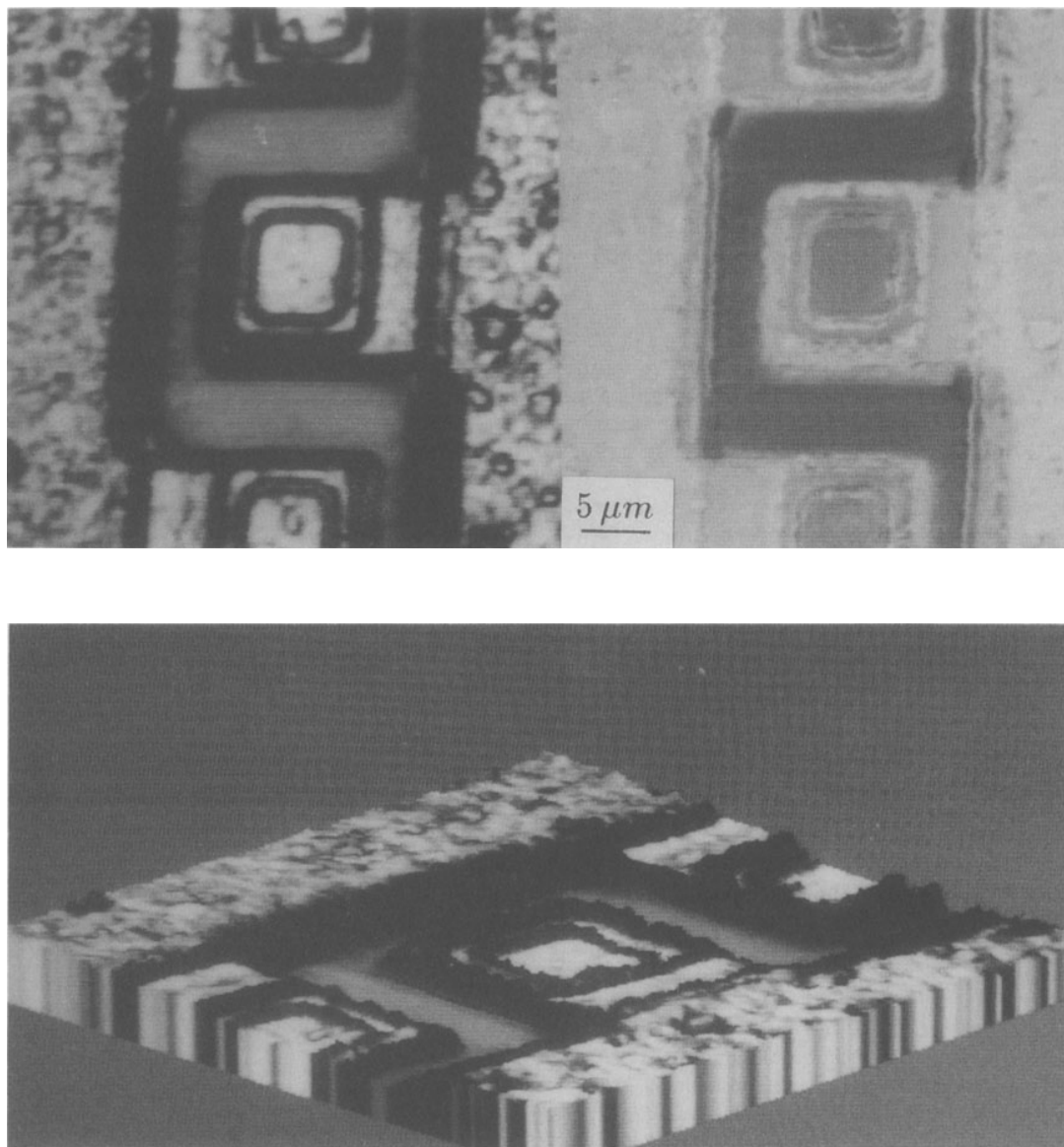


FIG. 1. An example of the kind of data processing available in confocal microscopy. The top left hand image shows an extended focus image of a portion of a particularly deep transistor. The top right hand image is the corresponding height image where object height is coded as image brightness. The bottom image is a computer generated isometric image of the same microcircuit.

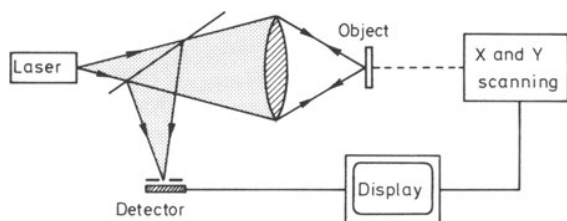


FIG. 2. The optical arrangement of a confocal scanning optical microscope.

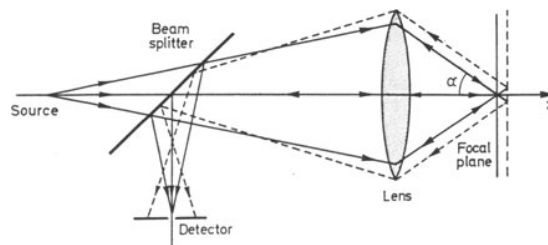


FIG. 3. The physical origin of the depth discrimination property.